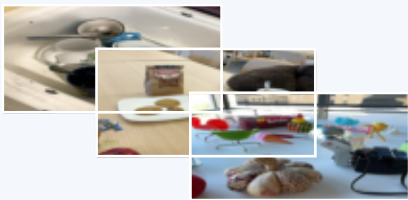


# CTF-GS: Compact Foundation-Feature Gaussian Memory

Frozen RADIO evidence is compressed into a compact 3D memory, then reconstructed features support 2D, 3D, and point queries.

## A. Multiview Evidence



posed RGB views + 3DGS geometry

**Frame-wise RADIO features**  
frozen 1280-D dense target

**Multiview primitive registration**  
visibility, contribution, confidence

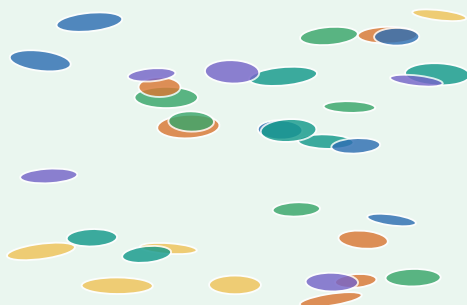
**Training constraints**  
feature, support, boundary, topology

compress

## B. Compact Scene Memory

### Compact Foundation-Feature Memory

stored once per scene; reconstruct features on demand



**compact code  $z_i$  + spatial context  $h(x)$**   
not explicit 1280-D feature storage

training only

## C. Open-Vocabulary Queries

### LERF rendered-view OVS

dense score map -> 2D mask



reconstruct

### LERF direct 3D OVS

primitive scores -> selected mask



calibrate

### ScanNet point query

open-vocabulary points



query

### Frozen-head probes

adaptor-space usability tests



probe

—> inference

- - - training only